
Chapter 9

Linking Spatial Exposure to Health Events

9.2 Linear Regression Models for Spatially Autocorrelated Data

A basic approach for the analysis of spatial data in the presence of a large-scale trend is to assume that all variation in the outcome variable of interest can be completely described by a collection of spatially varying covariates.

- This is just basic linear regression using the spatial locations as explanatory variables.

We will now consider the setting where there is a linear trend in the data, but the data are still spatially autocorrelated.

- The covariance matrix of the observed data $\mathbf{y}$ is $\text{Var}(\mathbf{y}) = \text{Var}(\epsilon) = \Sigma$.
- The (i, j) element of Σ is $\text{Cov}\left(Y(s_i), Y(s_j)\right)$.

Our basic model for the data in this setting is that

$$Y(s) = x(s)^T \beta + \epsilon(s),$$

where $x(s)$ is a vector of spatially varying covariates associated with location s , β is a set of regression coefficients, and $\epsilon(s)$ is a mean zero spatial random field.

$x(s)^T \beta$ (or $X\beta$ for the set of N responses) is referred to as the **mean structure, large-scale variation, or trend**.

The error term $\epsilon(s)$ (or ϵ for the set of N responses) defines the **small-scale variation** in the data and determines the **stochastic dependence structure or residual autocorrelation** of the data.

9.2.1 Estimation and Inference

If Σ is known, we may use maximum likelihood methods to obtain an estimator of β .

- It never is, except for simulated data.

Computationally, it is more convenient to assume that $\Sigma = \sigma^2 V$, where σ^2 is an unknown parameter to be estimated and the elements of V reflect the residual spatial correlation.

Assuming that $\mathbf{y} \sim MVN(X\beta, \sigma^2 V)$, the log likelihood is

$$\begin{aligned}\mathcal{L} &= \ln[l(\beta, \sigma^2 | \mathbf{y}, X)] \\ &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2} \ln|V| - \frac{(\mathbf{y} - X\beta)' V^{-1} (\mathbf{y} - X\beta)}{2\sigma^2},\end{aligned}$$

where $|V|$ is the determinant of V .

Differentiating with respect to these two parameters, and setting the derivatives equal to 0, we have

$$\frac{\partial \mathcal{L}}{\partial \beta} = \frac{1}{\sigma^2} (X'V^{-1}\mathbf{y} - X'V^{-1}X\beta) = 0$$

and

$$\frac{\partial \mathcal{L}}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} (\mathbf{y} - X\beta)'V^{-1}(\mathbf{y} - X\beta) = 0,$$

leading to the **generalized least squares (GLS) estimator**

$$\hat{\beta}_{GLS} = (X'V^{-1}X)^{-1}(X'V^{-1}\mathbf{y}),$$

and the maximum likelihood estimator of σ^2 , adjusted for the loss in degrees of freedom resulting from estimating β (i.e., the Restricted Maximum Likelihood (REML) estimator of σ^2), becomes

$$\hat{\sigma}_{REML}^2 = \frac{(\mathbf{y} - X\hat{\beta}_{GLS})' V^{-1} (\mathbf{y} - X\hat{\beta}_{GLS})}{N - p}.$$

Since Σ is unknown, and the number of free parameters to estimate in Σ is $N(N - 1)$, we have a major problem unless we parameterize Σ (like in the previous chapter).

Note that σ_{ML}^2 divides by N instead of $N - p$.

Assuming second-order stationary of the spatial process, we know that $C(s_i - s_j) = C(0) - \gamma(s_i - s_j)$.

- Using parametric covariance functions to describe Σ , we rewrite Σ as $\Sigma(\theta)$ to denote its dependence on the parameters $\theta = (a, c, c_0)$.

But we have a problem!

- Our data is NOT second-order stationary since we are now assuming it has a trend.
 - i.e., $E(Z(s)) \neq \mu$ for all $s \in D$.

We assume $\epsilon(s)$ is second-order stationary!

- But we still have a problem.

We can't estimate the semivariogram from the observed data!

Note that in our setting:

$$\begin{aligned} E[(Y(s_i) - Y(s_j))^2] &= \text{Var}(Y(s_i) - Y(s_j)) + [\mu(s_i) - \mu(s_j)]^2 \\ &= 2\gamma(s_i - s_j) + \left\{ \sum_{k=0}^p \beta_k [x_k(s_i) - x_k(s_j)] \right\}^2. \end{aligned}$$

What this means is that if we tried to estimate the semivariogram with the sample semivariogram of the observed data, we wouldn't be estimating the true semivariogram.

- We'd be estimating the true variogram plus something related to the trend of the data.

If we knew β , we could adjust our estimator to account for the bias.

But β is not known, and we need to know $\gamma(\cdot)$ or $\mathcal{C}(\cdot)$, to estimate it.

We have a chicken and egg type of problem!

We need a way to estimate β and $\gamma(\cdot)$ simultaneously.

We discuss two methods for this:

- 1. Iteratively Reweighted Generalized Least Squares (IRWGLS)**
- 2. Maximum Likelihood (ML)**

IRWGLS

After choosing a semivariogram model:

1. Obtain a starting estimate of β , say $\hat{\beta}$.
2. Compute residuals $r = \mathbf{y} - X\hat{\beta}$.
3. Estimate the semivariogram of the residuals using the sample semivariogram to get $\gamma(h, \hat{\theta})$ and $\Sigma(\hat{\theta})$.
4. Obtain a new estimate of β using $\hat{\beta} = \hat{\beta}_{GLS} = \left(X' \Sigma(\hat{\theta})^{-1} X\right)^{-1} X' \Sigma(\hat{\theta})^{-1} \mathbf{y}$.
5. Repeat steps 2-4 until the change in estimates of β and θ are small, i.e., the algorithm converges.

Note that estimates of the semivariogram parameters based on the residuals are biased, though the WRSS estimator is likely to suffer from smaller bias (hopefully).

IRWGLS gives the **estimated generalized least squares (EGLS) estimator**, $\hat{\beta}_{EGLS}$.

If we let $\hat{\theta}_{EGLS}$ denote the estimate of θ obtained from IRGLS, then the EGLS estimator of β is

$$\left(X' \Sigma(\hat{\theta}_{EGLS})^{-1} X\right)^{-1} X' \Sigma(\hat{\theta}_{EGLS})^{-1} \mathbf{y}$$

An estimator of the variance of $\hat{\beta}_{EGLS}$ is

$$\widehat{\text{Var}}(\hat{\beta}_{EGLS}) = \left[X' \Sigma(\hat{\theta}_{EGLS})^{-1} X \right]^{-1}.$$

If we denote the diagonal elements of this matrix by s^{kk} , the standard error for a single $[\hat{\beta}_{EGLS}]_k$ is $\sqrt{s^{kk}}$, and a $(1 - \alpha)100\%$ confidence interval for β_k is

$$[\hat{\beta}_{EGLS}]_k \pm \left(t_{N-p, \frac{\alpha}{2}} \right) \sqrt{s^{kk}},$$

where $t_{N-p, \frac{\alpha}{2}}$ is the $\alpha/2$ quantile of a t distribution with $N - p$ degrees of freedom.

Likelihood Methods

The log likelihood of our data is

$$\begin{aligned}\mathcal{L} &= \ln[l(\beta, \sigma^2 | \mathbf{y}, X)] \\ &= -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln|\Sigma(\theta)| - \frac{1}{2} (\mathbf{y} - X\beta)' \Sigma(\theta)^{-1} (\mathbf{y} - X\beta).\end{aligned}$$

The score equations for θ (first derivatives wrt θ set equal to 0) do not result in a closed-form solution for θ .

In practice, assuming an isotropic covariance model, we often parameterize our covariance function as

$$\Sigma(\theta) = \sigma^2 \left[R(a) + \frac{c_0}{\sigma^2} I_N \right] = \sigma^2 V(a, u)$$

where $R(a)$ is the isotropic correlation matrix of the observed data without a nugget (i.e., $\Sigma(\theta)$ with $\sigma^2 = 1$, $c_0 = 0$), σ^2 (denoted c in Chapter 8 by the book, but changed in Chapter 9), and $u = c_0/\sigma^2$.

In that case, we can instead maximize the **concentrated log likelihood** (or **profile likelihood**) obtained from substituting the solution to the score equations for β and σ^2 into the log likelihood above.

The profiled solutions for β and σ^2 , using the current values of a and u ($\tilde{a}$ and $\tilde{u}$) are

$$\tilde{\beta} = (X'[V(a, u)]^{-1}X)^{-1}(X'[V(a, u)]^{-1}\mathbf{y})$$

and

$$\tilde{\sigma}^2 = \frac{(\mathbf{y} - X\tilde{\beta})'[V(a, u)]^{-1}(\mathbf{y} - X\tilde{\beta})}{N}.$$

The concentrated log likelihood then only depends on (a, u) , and we seek to minimize

$$\mathcal{L}^*(a, u) = -\frac{1}{2} [N \ln 2\pi + N \ln \tilde{\sigma}^2 + \ln |V(a, u)| + N].$$

- There are many optimization algorithms that can be used to minimize $\mathcal{L}^*(a, u)$ with respect to (a, u)
- See the `optim` and `nlm` functions in R.

The asymptotic standard errors of the maximum likelihood estimators are obtained from the inverse of the information matrix (see the book for details).

The standard error and confidence intervals for $\tilde{\beta}_k$ can be obtained in a similar manner to before (see the book for details), using the asymptotic standard errors.

We can obtain confidence intervals for the elements of θ using

$$\tilde{\theta}_\ell \pm z_{\alpha/2} \sqrt{\widehat{\text{Var}}(\tilde{\theta}_\ell)},$$

where $\text{Var}(\tilde{\theta}_\ell)$ comes from the inverse of the information matrix, substituting ML estimates for any unknown parameters.

To perform a test comparing a set of nested models, $H_0: \theta = \theta_1$ against the alternative $H_a: \theta = \theta_2$, we use the statistic $2[\mathcal{L}(\beta, \theta_2) - \mathcal{L}(\beta, \theta_1)]$, and compare this statistic to a χ^2 distribution with $\dim(\theta_2) - \dim(\theta_1)$ degrees of freedom (assuming that the reduced model is nested in the larger).

- This assumes the models have the same form and covariates, but the parameters are different.

To compare two models that are not nested, we can use **Akaike's information criterion (AIC)**, a penalized log-likelihood ratio defined by

$$AIC(\tilde{\beta}, \tilde{\theta}) = -2\mathcal{L}(\tilde{\beta}, \tilde{\theta}) + 2(p + g),$$

where $p + g$ is the number of parameters in the model (p is the number of regression coefficients, g is the number of covariance parameters).

9.2.2 Interpretation and Use with Spatial Data

“One person’s trend structure is another person’s covariance”.

It can be difficult to separate our covariance effects from our spatial dependence.

- We may get very similar fits for two models that differ by a single covariate (and properly adjusted spatial autocorrelation).
- The interpretation of these will be very different.
- We must thoughtfully consider our model of the data in light of this, always being aware of the fact that our interpretations are (generally) made under the assumption that our fitted model structure is in fact the true one.

9.2.3 Predicting New Observations: Universal Kriging

Universal kriging is optimal linear prediction when the mean is unknown and nonstationary (i.e., not constant).

- The universal kriging predictor is unbiased and chosen to be the linear predictor that minimizes the mean square prediction error.

Let $\Gamma = \{\gamma(s_i - s_j)\}_{i,j=1}^N$ and $\gamma = \{\gamma(s_0 - s_i)\}_{i=1}^N$, where s_0 is a site of interest in the study area. Let x_0 be the set of covariates for s_0 .

Assuming intrinsic stationarity for the random field $\epsilon(s)$, the universal kriging predictor of $Y(s_0)$ is

$$\hat{Y}(s_0) = [\gamma + X(X'\Gamma^{-1}X)^{-1}(x_0 - X'\Gamma^{-1}\gamma)]'\Gamma^{-1}\mathbf{y} = \lambda'\mathbf{y}.$$

The associated mean square prediction error is

$$\sigma_k^2(s_0) = 2\lambda'\gamma - \lambda'\Gamma\lambda.$$

If this is the mean square prediction error for filtered kriging, then the formula becomes $\sigma_k^2(s_0) = 2\lambda'\gamma - \lambda'\Gamma\lambda - c_0$, where c_0 is the measurement error variance (nugget variance).

Let $C = \{C(s_i - s_j)\}_{i,j=1}^N$ and $c = \{C(s_0 - s_i)\}_{i=1}^N$, where s_0 is a site of interest in the study area.

If we assume second-order stationarity for the random field $\epsilon(s)$, the universal kriging predictor of $Y(s_0)$ is

$$\hat{Y}(s_0) = [c + X(X'C^{-1}X)^{-1}(x_0 - X'C^{-1}c)]'C^{-1}\mathbf{y} = \lambda'\mathbf{y}.$$

The associated mean square prediction error is

$$\sigma_k^2(s_0) = C(0) - 2\lambda'c + \lambda'c\lambda.$$

It may be tempting to bypass universal kriging when your data has a trend, and instead apply ordinary kriging to the detrended data.

- i.e., apply ordinary kriging to $\hat{\epsilon} = \mathbf{y} - X\hat{\beta}$.
- Your prediction of $Y(s_0)$ becomes $x_0^T \hat{\beta} + \hat{\epsilon}_{OK}$

Surprisingly, this will result in identical predicted values to universal kriging if the covariance/semivariograms used are the same in both settings and $\hat{\beta}$ is the resulting GLS estimator of β .

However, the kriging variance for the detrended data will be too small in comparison to the universal kriging variance because the detrended data prediction doesn't account for the fact that β is estimated, while universal kriging does.

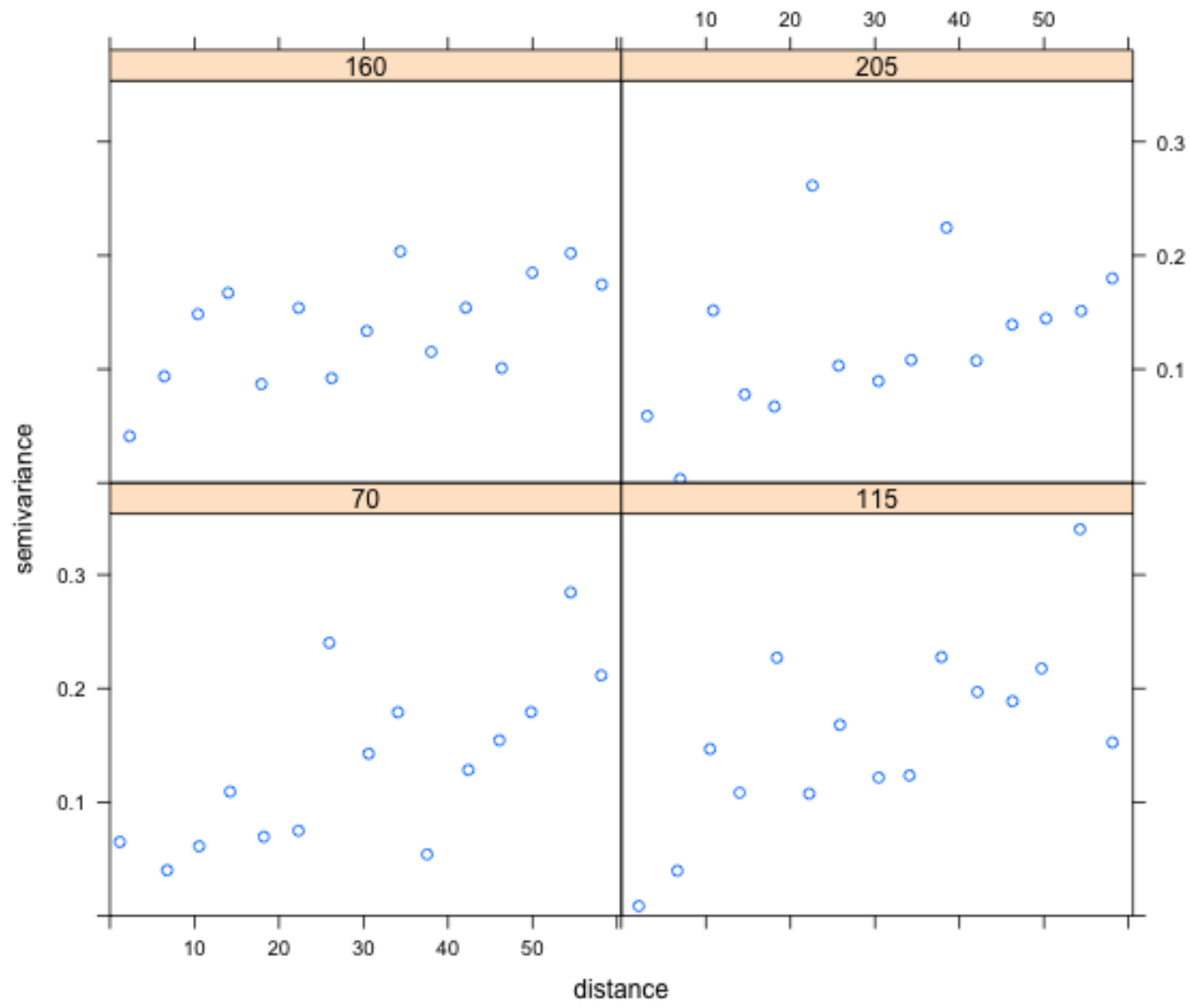
A simple (but not optimal) approach to estimating the semivariogram in a universal kriging context is to use ordinary least squares to estimate the regression coefficients, then estimate the semivariogram using the ordinary least squares residuals.

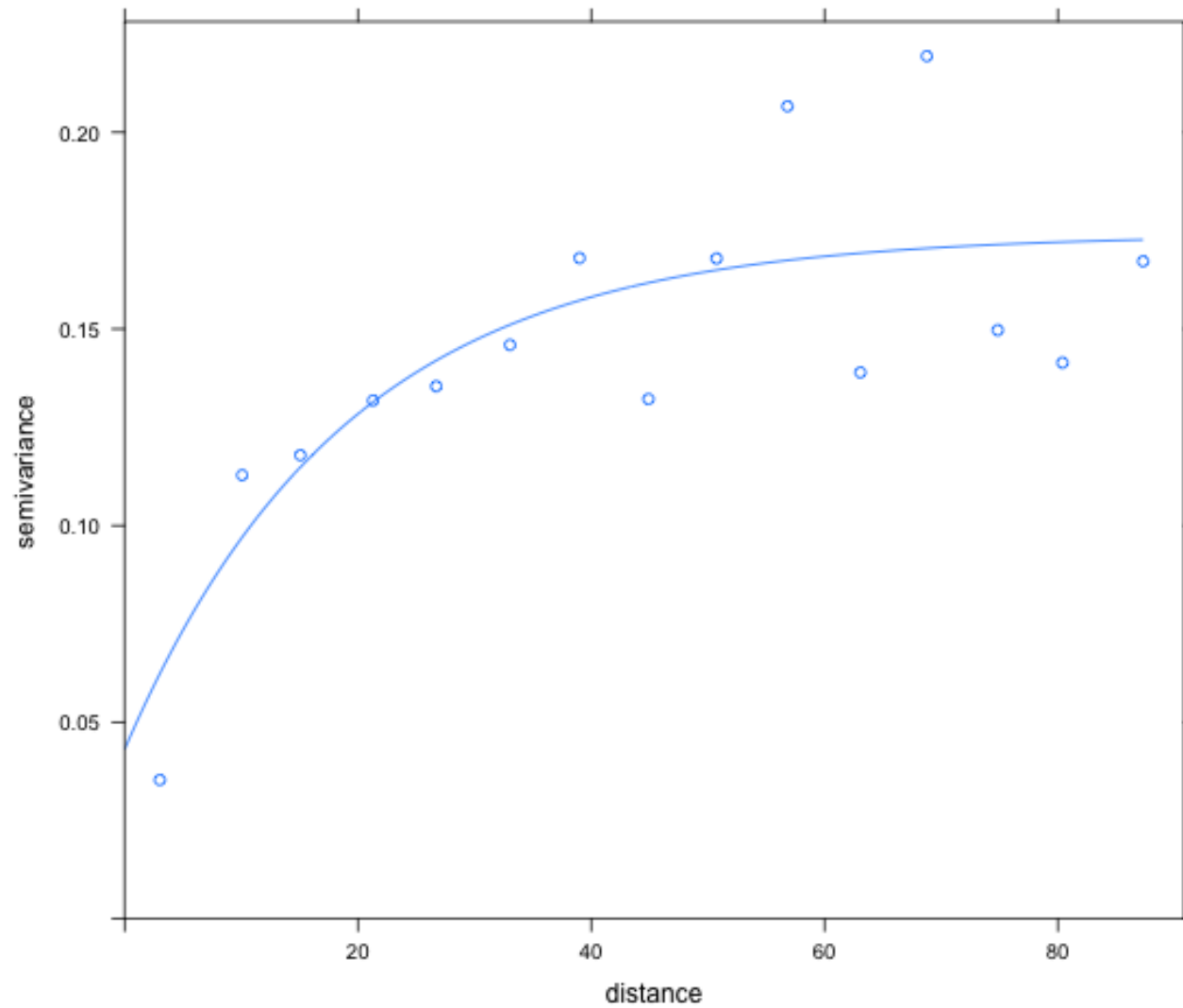
- This is what we will do for the purpose of illustration.

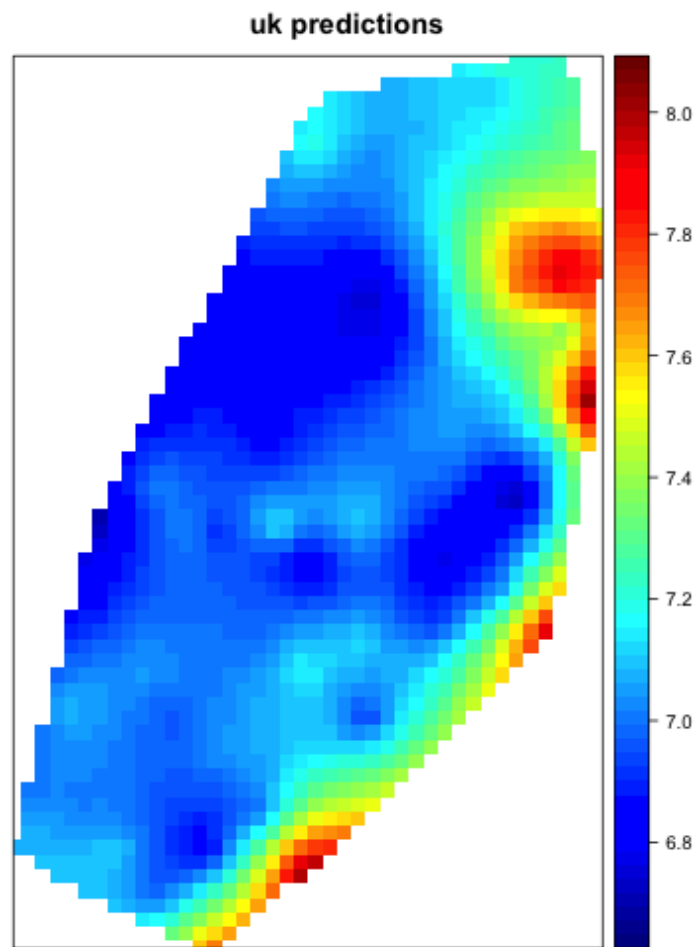
Data Break continued:

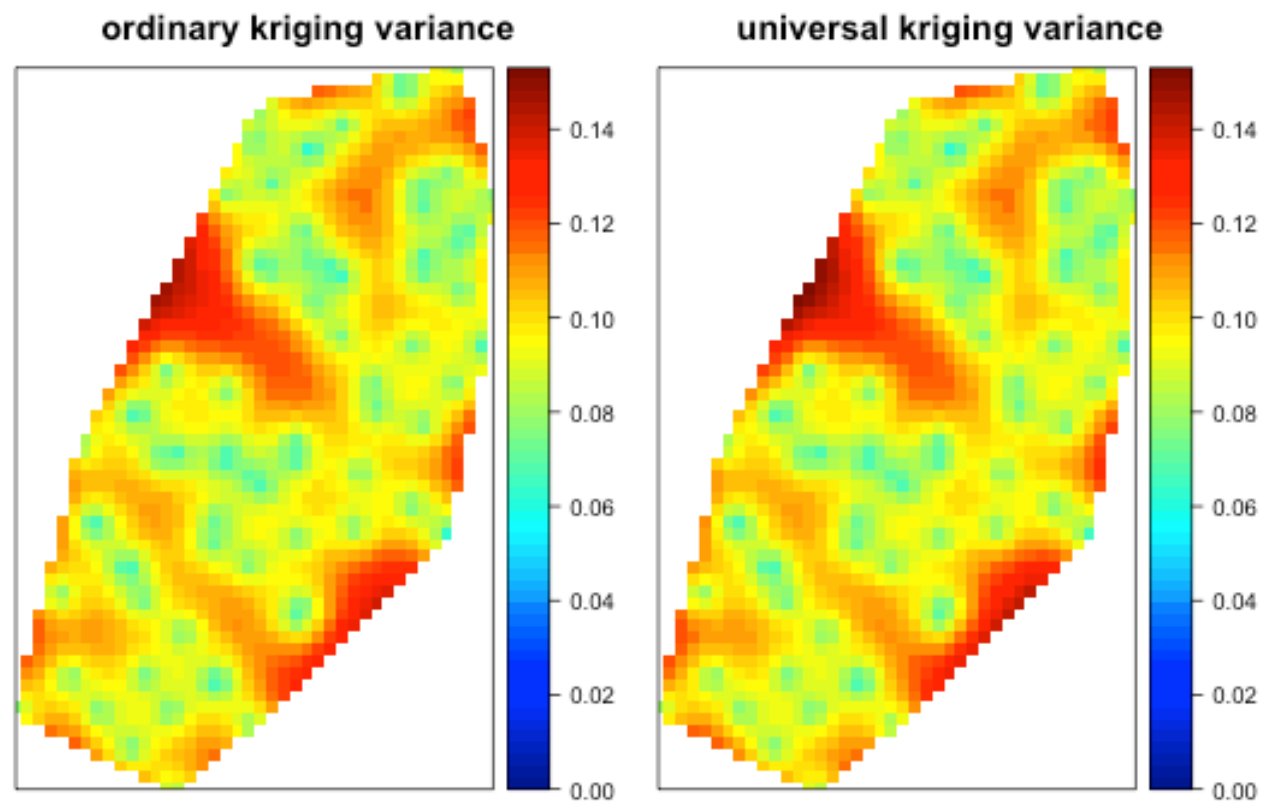
We estimate the semivariogram and make predictions in a universal kriging context.

- The directional variogram show no real signs of anisotropy, so we proceed with estimation of an isotropic semivariogram/covariance.









9.3 Spatial Autoregressive Models

Once we define the covariates of our regression model, a spatial regression model is completed by specifying the elements of the variance-covariance matrix Σ .

- For geostatistical data, this is done in terms of distance.
- For areal data, this is done in terms of spatial proximity matrices since we restrict inference to the set of observed regions

Instead of modeling the spatial covariance structure directly for areal data, we induce it by regressing the response on each other.

We use the term **autoregressive** to reflect self-regression.

9.3.1 Simultaneous Autoregressive Models

We apply the idea of spatial autoregression to the vector of residuals errors, ϵ , i.e., we regress one residual on the others through the formula

$$\epsilon(s_i) = \sum_{j=1}^N b_{ij} \epsilon(s_j) + v(s_i),$$

with $b_{ii} = 0$ so we don't regress a residual on itself, as assume that the residual errors ("residual residuals") from this regression, $v(s_i)$, $i = 1, \dots, N$, have mean zero and a diagonal variance-covariance matrix $\Sigma_v = \text{diag}[\sigma_1^2, \dots, \sigma_N^2]$.

The $\{b_{ij}\}$ terms represent **spatial dependence parameters** since they measure the contributions of the "other" observation to the variation of $\epsilon(s_i)$.

Our linear regression model $Y(s_i) = x(s_i)' \beta + \epsilon(s_i)$, where $x(s_i) = (1, x_1(s_i), \dots, x_{p-1}(s_i))$, becomes

$$Y(s_i) = x(s_i)' \beta + \sum_{j=1}^N b_{ij} \epsilon(s_j) + v(s_i).$$

If all the b_{ij} are zero, the model reduces to the traditional linear regression model with independent errors.

Using the relationship $\epsilon(s_j) = Y(s_j) - x(s_j)' \beta$ from our original linear regression model, we obtain

$$Y(s_i) = x(s_i)' \beta + \sum_{j=1}^N b_{ij} [Y(s_j) - x(s_j)' \beta] + v(s_i), i = 1, \dots, N.$$

Representing the model in this way allows us to assess the degree of spatial dependence in the data through the inclusion of the term $\sum_{j=1}^N b_{ij} \left[Y(s_j) - x(s_j)' \beta \right]$ (a weighted sum of the deviation of the j th observation from its model mean value) while controlling for the variate effects within region i .

Conversely, the formula allows us to measure covariate effects while controlling for spatial dependence.

The model can be expressed in matrix form as

$$(I - B)(\mathbf{y} - X\beta) = \mathbf{v},$$

where the $N \times N$ matrix B contains the spatial dependence parameters, b_{ij} , and $b_{ii} = 0$ for all i .

The variance-covariance matrix of $\mathbf{y}$ is

$$\Sigma_{\mathbf{y}} = \text{Var}(\mathbf{y}) = (I - B)^{-1} \Sigma_v (I - B')^{-1},$$

if $(I - B)^{-1}$ exists.

The parameters b_{ij} induce the variance-covariance matrix of $\mathbf{y}$.

Spatial Dependence and Covariance Matrices

To simplify inference, we parameterize B as $B = \rho W$, where W is a spatial proximity matrix defined previous and ρ is a parameter to be estimated.

In this setting, our SAR model can be written as

$$Y(s_i) = x(s_i)' \beta + \rho \sum_{j \in \mathcal{N}_i} w_{ij} [Y(s_j) - x(s_j)' \beta] + v(s_i),$$

for $i = 1, \dots, N$, where N denotes the number of regions and $\mathcal{N}_i$ denotes the neighborhood set of i th region.

In matrix form, we often write this as

$$\begin{aligned}\mathbf{y} &= X\beta + (I - \rho W)^{-1}v \\ &= X\beta + \rho W\mathbf{y} - \rho WX\beta + v \\ &= X\beta + \rho W(\mathbf{y} - X\beta) + v.\end{aligned}$$

Notice that we now have two additional terms in our regression model, $\rho WX\beta$ and $\rho W\mathbf{y}$.

- These are **spatially lagged variables**.
- In a SAR model, both the covariates and response are spatially lagged.

For $(I - \rho W)$ to be invertible, the eigenvalues of W have to satisfy certain constraints (see the book for details).

Estimating the Parameters in SAR Models

The parameters in SAR models are estimated via maximum likelihood-based methods.

We typically reparameterize Σ_v as $\Sigma_v = \sigma^2 V_v$, so the covariance matrix of the SAR model becomes

$$\Sigma_{SAR} = \sigma^2 (I - B)^{-1} V_v (I - B')^{-1} = \sigma^2 V_{SAR}(\theta),$$

where θ contains the b_{ij} and the parameters in V_v .

The details of estimating the parameters is complicated.

- See the book for details.
- The fitting can be done straightforwardly in R.

Testing Residual SAR Dependence

The one-parameter SAR model ($B = \rho W$ and $\Sigma_v = \sigma^2 I$) is by far the most prevalent SAR model used in practice.

We can use this to test whether $\rho = 0$ for the model $\mathbf{y} = X\beta + (I - \rho W)^{-1}v$.

- If $\rho = 0$, this simplifies to the standard independent error regression model.

Moran's I with OLS residuals

We can assess for spatial autocorrelation among the residuals of our linear model by modifying Moran's I slightly.

We replace the overall mean $\bar{Y}$ in Moran's I, with the model's predicted value for each observation $\hat{Y}_i = x(s_i)^T \hat{\beta}$.

Under the assumption that the responses are multivariate normal and independent of each other, the asymptotic mean and variance of the Moran's I statistic can be calculated, from which a standardized test statistic and associated p-value can be determined.

The interpretation of the Moran's I, after correcting for the mean, is essentially the same as before.

Note that Moran's I doesn't tell you the type of deviation from the null hypothesis, i.e., it doesn't tell you how the responses deviate from independence (it could be in terms of a SAR model, or it could be something else).

See the book for numerous additional details.

Wald Test

Since the maximum likelihood estimators are asymptotically Gaussian, an approximate confidence interval for ρ is

$$\hat{\rho} \pm z_{\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\rho})}.$$

A Wald test of $H_0: \rho = 0$ is made by comparing $z = \hat{\rho} / \sqrt{\widehat{\text{Var}}(\hat{\rho})}$ with the quantiles of a standard normal distribution.

Likelihood Ratio Test

Suppose we want to compare two models with $\theta_1 = (\beta', \sigma^2)'$ to $\theta_2 = (\beta', \sigma^2, \rho)'$.

A test of $H_0: \theta = \theta_1$ versus the alternative $H_a: \theta = \theta_2$ (equivalent to $H_0: \rho = 0$ versus $H_a: \rho \neq 0$) can be performed using the statistic $2(\mathcal{L}(\beta, \sigma^2, \rho) - \mathcal{L}(\beta, \sigma^2, 0))$ to a χ^2 distribution with $\dim(\theta_2) - \dim(\theta_1) = 1$ degrees of freedom.

Concluding Remarks

The tests mentioned above were derived using asymptotic arguments, which may not be appropriate.

Monte Carlo hypothesis testing isn't as simple as before and is much more labor intensive.

Other alternatives are available.

All tests discussed were applied to OLS residuals.

- Appropriate tests of spatial autocorrelation in error terms of SAR models (or any other spatial models for that matter) remains an open area of research.

9.3.2 Conditional Autoregressive Models

In the previous sections (9.2 and 9.3.1), we presented models of the **joint** probability distribution $f(Y(s_1), \dots, Y(s_N))$ with spatial association incorporated through the variance-covariance matrix $\Sigma = \text{Var}(\mathbf{y})$.

In some settings, it is more convenient to model the set of **conditional** probability distributions of each observation, $Y(s_i)$, given the observed values of all other observations.

- We model $f(Y(s_i) \mid \mathbf{y}_{-i})$, where $\mathbf{y}_{-i}$ is the vector of all observations except the current one.

We once again assume $Y(s_i)$ only depends on a set of **neighbors**.

By assuming the responses are Gaussian, we only need to specify the mean and conditional variance of each observation to complete our model.

A **Conditional Autoregressive (CAR) model** is defined by specifying the conditional mean and variance:

$$E[Y(s_i) | \mathbf{y}_{-i}] = x(s_i)' \beta + \sum_{j=1}^N c_{ij} [Y(s_j) - x(s_j)' \beta],$$

$$\text{Var}[Y(s_i) | \mathbf{y}_{-i}] = \sigma_i^2, \quad i = 1, 2, \dots, N,$$

where the c_{ij} denote spatial dependence parameters.

The c_{ij} are only nonzero if $j \in \mathcal{N}_i$ and $c_{ii} = 0$.

The **Hammersly-Clifford theorem** (Besag 1974) describes the conditions necessary for a set of conditional distributions to define a valid joint distribution.

- It's straightforward to derive conditional Gaussian distributions from a multivariate Gaussian joint distribution.
- It's less straightforward to go the other direction.

A set of conditional distributions defined over spatial neighborhoods and meeting these conditions defined a **Markov random field**, where each observation given the other observations only depends on values at neighboring locations.

The conditional means and variances generate a valid multivariate Gaussian distribution with mean $X\beta$ and variance-covariance matrix

$$\Sigma_Y = (I - C)^{-1} \Sigma_c,$$

where $\Sigma_c = \text{diag}[\sigma_1^2, \dots, \sigma_N^2]$.

- The condition $\sigma_j^2 c_{ij} = \sigma_i^2 c_{ji}$ must be satisfied for all i and j for the variance-covariance matrix to be symmetric.

Relationship to SARs

If we set $\Sigma_c = \sigma^2 I$, and in the development of a SAR model, set $\Sigma_v = \sigma^2 I$, then any SAR model can be written as a CAR model with $C = B + B' - BB'$.

Any CAR model can be written as a SAR model, but the relationships are contrived, and the neighborhood structure of the models may not be the same.

Estimation and Inference with CAR Models

We parameterize Σ_c as $\Sigma_c = \sigma^2 V_c$, so the variance-covariance matrix of a CAR model can be written as

$$\Sigma_{CAR} = \sigma^2 (I - C)^{-1} V_c = \sigma^2 V_{CAR}(\theta),$$

where θ contains the c_{ij} and the elements of V_c .

Since we can write the joint likelihood of our model, we can proceed with likelihood-based estimation and inference methods, as with SAR models.

In the one-parameter CAR model, we choose $C = \rho W$.

9.3.3 Concluding Remarks on Conditional Autoregressions

The primary purpose of CAR models is to provide a modeling mechanism accounting for residual spatial correlation (not explained by spatial patterns in covariate values).

Spatial autoregressive models were derived for areal data.

Prediction of new observations is typically not a focus of SAR or CAR models.

When data are missing, prediction of areal values may be a goal, and in such cases, the universal kriging predictor can be used, with Σ_Y corresponding to the one coming from a SAR or CAR model.

DATA BREAK: New York Leukemia DATA (*cont.*)

We have already used the New York state eight-county census tract data set for examining the construction of neighbor lists and spatial weights.

There are 281 census tract observations, including as we have seen sparsely populated rural areas contrasting with dense, small, urban tracts.

The numbers of incident leukemia cases are recorded by tract, aggregated from census block groups, but because some cases could not be placed, they were added proportionally to other block groups, leading to non-integer counts.

The counts are for the 5 years 1978–1982, while census variables, such as the tract population, are just for 1980.

Other census variables are the percentage aged over 65, and the percentage of the population owning their own home.

Exposure to trichloroethylene (TCE) waste sites is represented as the logarithm of 100 times the inverse of the distance from the tract centroid to the nearest site. We return to these covariates in following sections.

We consider further analysis of the New York leukemia data. To make the counts of leukemia cases more closely follow a Gaussian distribution, we make the transformation

$$Z_i = \log \frac{1000(Y_i + 1)}{n_i},$$

where Y_i is the rate of leukemia cases in region i and n_i is the number of population size of region i .

The transformed incidence proportions are not yet normal, with three outliers, tracts with small populations but unexpectedly large case counts.

They could be smoothed away, but may in fact be interesting, as the patterns they display may be related to substantive covariates, such as closeness to TCE locations.

As covariates, we have used the inverse distance to the closest TCE (PEXPOSURE), the proportion of people aged 65 or higher (PCTAGE65P) and the proportion of people who own their own home (PCTOWNHOME).

We'll explore how accounting for covariates, spatial dependence, and different models affect things.

We assume the data follow a multivariate Gaussian distribution with mean independent of variance.

The incidence proportions DO NOT satisfy these assumptions, but we'll ignore this.

We will check for residual spatial autocorrelation in OLS, SAR, and CAR models, as well as doing some smoothing.

We use a binary connective matrix.

9.3.4 Conclusion Remarks on Spatial Autoregressions

Accounting for population heterogeneity and incorporating spatial correlation can be crucial for making proper inference.

Spatial autoregressive models incorporate spatial correlation through the spatial proximity matrices.

- Geostatistical approaches could be used, utilizing intercentroid distances.

Differences in spatial proximity measures can lead to very different results.

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1. It is better to use a reasonable method for modeling spatial autocorrelation than to wrongly assume the data are independent.
 2. The choice of spatial dependence can greatly affect estimated parameter values and standard errors.
 - a. Explore your data before deciding on a spatial dependence model.
 - b. Evaluate any differences between resulting models.
 3. Asymptotic results are justified by having lots of data. Make sure you have enough. (Divide the number of observations by 2, e.g.)
 4. Accounting for population heterogeneity is very important.
 5. Choose the simplest model that adequately explains the data.